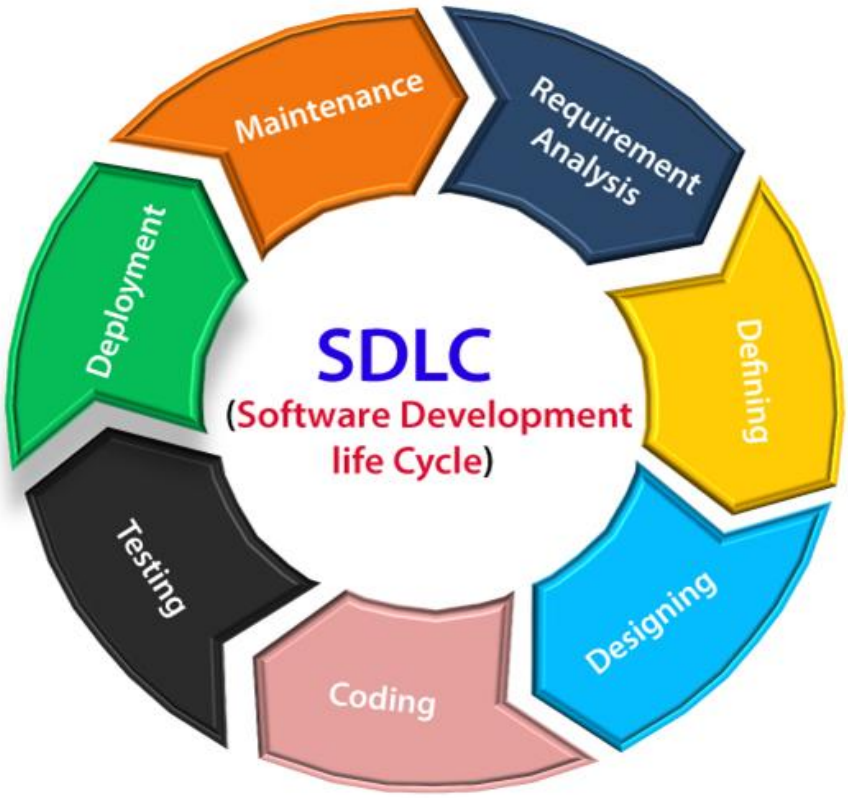
1.SOFTWARE DEVELOPMENT LIFE CYCLE (SDLC):

Software Development Life Cycle (SDLC) is a structured process that is used to design, develop and test good-quality software. SDLC or Software Development Life Cycle is a methodology that defines the entire procedure of software development step-by-step.

The goal of Software Development Life Cycle (SDLC) model is to deliver high-quality, maintainable software that meets the user's requirements.

In other words, life cycle model maps the various activities performed on a software product from its inception to retirement.

The SDLC defines and outlines a detailed plan with stages, or phases, that each encompass their own process and deliverables. The SDLC enhances development speed and minimizes project risks and costs associated with alternative methods of production.



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1.1 Requirement Analysis:

- Requirement analysis is the most important and fundamental stage in SDLC. It is performed by the senior members of the team with inputs from the customer, the sales department, market surveys and domain experts in the industry. This information is then used to plan the basic project approach and to conduct product feasibility study in the economical, operational and technical areas.
- Planning for the quality assurance requirements and identification of the risks associated with the project is also done in the planning stage. The outcome of the technical feasibility study is to define the various technical approaches that can be followed to implement the project successfully with minimum risks.

1.2 Defining:

- Once the requirement analysis is done the next step is to clearly define and document the product requirements and get them approved from the customer or the market analysts.
- This is done through a Software Requirement Specification (SRS) document which consists of all the product requirements to be designed and developed during the project life cycle.

1.3 Designing:

- After the completion of the Defining stage the next step is Designing of the project. In the design phase, software engineers analyse requirements and identify the best solutions to create the software. This phase is the product of the last two, like inputs from the customer and requirement gathering.
- A design approach clearly defines all the architectural modules of the product along with its communication and data flow representation with the external and third-party modules (if any).
- The system design phase includes creating the software's architecture and design. Based on the requirements gathered during planning, the team creates a blueprint outlining how the software will function.

1.4 Coding:

- In the coding or developing phase of SDLC, the actual development begins, and the programming is built. The implementation of design begins concerning writing code.
- Developers must follow the coding guidelines defined by their organization and programming tools like compilers, interpreters, debuggers, etc. are used to generate the code. Different high level programming languages such as C, C++, Pascal, Java and PHP are used for coding. The programming language is chosen with respect to the type of software being developed.

1.5 Testing:

- After the code is generated, it is tested against the requirements to make sure that the products are solving the needs addressed and gathered during the requirements stage.
- However, this stage refers to the testing only stage of the product where product defects are reported, tracked, fixed and retested, until the product reaches the quality standards defined in the SRS.

1.6 Deployment:

- Once the software is certified, and no bugs or errors are stated, then it is deployed.
- Then based on the assessment, the software may be released as it is or with suggested enhancement in the object segment.
- In this phase, the actual coding and development of the system take place. Developers build the system according to the design specifications, implementing features, creating databases, and writing code.
- After the software is deployed, then its maintenance begins.

1.7 Maintenance:

- Maintenance is an ongoing phase where the system is monitored, maintained, and updated as needed. This includes bug fixes, performance enhancements, security patches, and responding to user feedback.
- Proper maintenance ensures the system remains efficient, secure, and adaptable to future business needs.

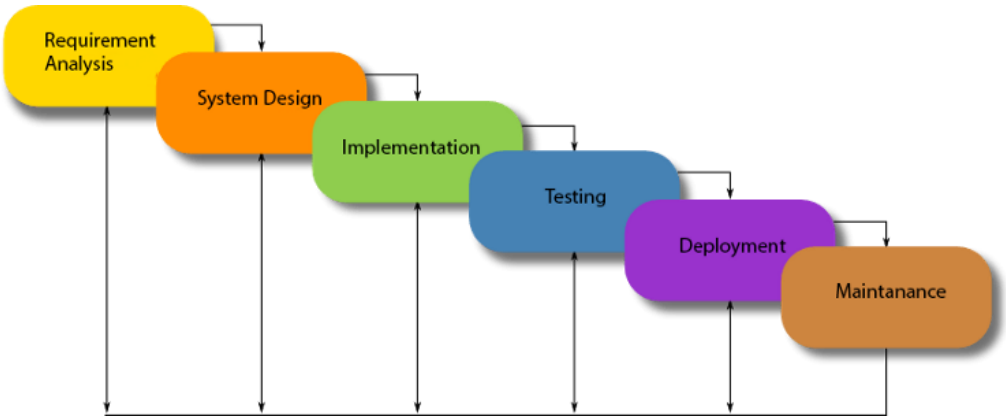
Software Development Life Cycle Models:

- Waterfall Model
- Iterative Model
- Evolutionary Model
- Prototype Model
- Spiral Model
- RAD Model
- Agile Model
- Incremental Model

Waterfall Model:

- The Waterfall Model was the first Process Model to be introduced. It is also referred to as a linear-sequential life cycle model. It is very simple to understand and use. In a waterfall model, each phase must be completed before the next phase can begin and there is no overlapping in the phases.
- The Waterfall model is the earliest SDLC approach that was used for software development.
- The waterfall Model illustrates the software development process in a linear sequential flow. This means that any phase in the development process begins only if the previous phase is complete. In this waterfall model, the phases do not overlap.
- Waterfall approach was first SDLC Model to be used widely in Software Engineering to ensure success of the project.

- In "The Waterfall" approach, the whole process of software development is divided into separate phases.
- In this Waterfall model, typically, the outcome of one phase acts as the input for the next phase sequentially.



- Requirement Gathering and analysis – All possible requirements of the system to be developed are captured in this phase and documented in a requirement specification document.
- System Design – The requirement specifications from first phase are studied in this phase and the system design is prepared. This system design helps in specifying hardware and system requirements and helps in defining the overall system architecture.
- Implementation – With inputs from the system design, the system is first developed in small programs called units, which are integrated in the next phase. Each unit is developed and tested for its functionality, which is referred to as Unit Testing.
- Integration and Testing – All the units developed in the implementation phase are integrated into a system after testing of each unit. Post integration the entire system is tested for any faults and failures.
- Deployment of system – Once the functional and non-functional testing is done; the product is deployed in the customer environment or released into the market.
- Maintenance – There are some issues which come up in the client environment. To fix those issues, patches are released. Also to enhance the product some better versions are released. Maintenance is done to deliver these changes in the customer environment.

2. DATA DICTIONARY:

2.1 admin

This table is storing the information of the Admin. It stores id, username and password of the admin which is used for login to the system.

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTION
a_id	int	6	Primary Key	Admin Id
a_username	varchar	20	Not Null	Admin Name
a_password	varchar	15	Not Null	Admin Password

2.2 faculty

This table is storing the information of the Faculty. It stores id, username and password of the faculty which is used for login to the system.

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTION
f_id	int	6	Primary Key	Faculty Id
f_username	varchar	20	Not Null	Faculty Name
f_password	varchar	15	Not Null	Faculty Password

2.3 student

This table is storing the information of the student. It stores id, username and password of the student which is used for login to the system.

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTION
s_id	int	6	Primary Key	Student Id
s_username	varchar	20	Not Null	Student Name
s_password	varchar	15	Not Null	Student Password

2.4 courses

This table is storing the information of the courses. It stores id, name and Description of the courses.

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTION
c_id	int	6	Primary Key	Course Id
c_name	varchar	20	Not Null	Course Name (Ex. BCA, MCA)
description	varchar	150	Not Null	Course Description

2.5 notice

This table is storing the information of the notice. It stores id, message and admin id.

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTION
id	int	6	Primary Key	Notice Id
message	varchar	200	Not Null	Message for the students
date	date		Not Null	Date of Notice

2.6 Feedback

This table is storing the information of the feedback. It stores id of feedback and message.

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTION
id	int	6	Primary Key	Feedback Id
s_id	int	6	Foreign Key	Student Id
feedback	varchar	200	Not Null	Enter Feedback

2.7 Alumni

This table is storing the information of the Alumni. It stores id of alumni, name of alumni, passing year, company, position and e-mail.

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTION
id	int	6	Primary Key	Alumni Id
name	varchar	20	Not Null	Alumni Name
passing_year	varchar	8	Not Null	Alumni Passing year
company	varchar	25	Not Null	Alumni Company
position	varchar	20	Not Null	Alumni Position
e-mail	varchar	20	Not Null	Alumni E-Mail

2.8 e_learning_course

This table is storing the information of the E-Learning Courses. It stores Id of E-Learning Course, Name of E-Learning Course and Id of Faculty.

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTION
id	int	6	Primary Key	Id of E-Learning Course
f_id	int	6	Foreign Key	Faculty Id
name	varchar	20	Not Null	Name of E-Learning Course

2.9 s_leave

This table is storing information of the student leave.

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTION
id	int	6	Primary Key	Leave Id
s_id	int	6	Foreign Key	Student Id
s_name	varchar	20	Not Null	Student Name
s_date	date		Not Null	Starting Date
e_date	date		Not Null	Ending Date
reason	varchar	50	Not Null	Reason of Leave
approve	varchar	10	Not Null	Display leave approve or not

2.10 f_leave

This table is storing information of the faculty leave.

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTION
id	int	6	Primary Key	Leave Id
f_id	int	6	Foreign Key	Faculty Id
f_name	varchar	20	Not Null	Faculty Name
s_date	date		Not Null	Starting Date
e_date	date		Not Null	Ending Date
reason	varchar	50	Not Null	Reason of Leave
approve	varchar	10	Not Null	Display leave approve or not

2.11 subject

This table is storing information of the Subject. It stores id, subject code, subject name, Course name and Faculty Name.

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTION
id	int	6	Primary Key	Subject Id
sub_code	int	5	Not Null	Subject Code
sub_name	varchar	15	Not Null	Subject Name
course_name	varchar	15	Foreign Key	Course Name
f_username	varchar	20	Foreign Key	Faculty Name

2.12 assignment

This table is storing information of the assignment. It stores id, subject name, submission date, Faculty id and Assignment Pdf.

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTION
id	int	6	Primary Key	Assignment Id
sub_name	varchar	15	Foreign Key	Subject Name
submission_date	date		Not Null	Submission Date
f_id	int	6	Foreign Key	Faculty Id
ass_pdf	varchar	15	Not Null	Assignment PDF

2.13 attendance

This table is storing information of the Attendance. It stores Id, Date, Fill and Faculty id.

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTION
id	int	6	Primary Key	Attendance Id
s_id	int	6	Foreign Key	Student Id
date	date		Not Null	Date of Attendance
fill	varchar	5	Not Null	Present, Absent
f_id	int	6	Foreign Key	Faculty Id

2.14 admission

FIELD NAME	DATATYPE	SIZE	CONSTRAINTS	DESCRIPTION
id	int	6	Primary Key	Admission Id
first_name	varchar	15	Not Null	First Name of Student
last_name	varchar	15	Not Null	Last Name of Student
father_name	varchar	15	Not Null	Father Name of Student
mother_name	varchar	15	Not Null	Mother Name of Student
gender	varchar	6	Not Null	Gender of Student
dob	date		Not Null	Birth Date of Student
phone_no	varchar	10	Not Null	Phone No of Student
email_id	varchar	20	Not Null	E-mail Id of Student
f_phone_no	varchar	10	Not Null	Father's Phone No
address	varchar	50	Not Null	Address
category	varchar	15	Not Null	Category of Student (Ex. OBC, SC)
city	varchar	15	Not Null	City of Student
disability	varchar	5	Not Null	Disability of Student
f_occupation	varchar	20	Not Null	Father's Occupation
stud_photo	varchar	15	Not Null	Student Photo
aadhar_card	varchar	15	Not Null	Student Aadhar Card
10_result	varchar	15	Not Null	10th Result of Student
12_result	varchar	15	Not Null	12th Result of Student
leaving_certificate	varchar	15	Not Null	Leaving Certificate

3. DATA FLOW DIAGRAM (DFD):

A data flow diagram (DFD) maps out the flow of information for any process or system. It uses defined symbols like rectangles, circles and arrows, plus short text labels, to show data inputs, outputs, storage points and the routes between each destination.

Data flowcharts can range from simple, even hand-drawn process overviews, to in-depth, multi-level DFDs that dig progressively deeper into how the data is handled. They can be used to analyse an existing system or model a new one.

Data Flow Diagram (DFD) represents the flow of data within information systems. Data Flow Diagrams (DFD) provide a graphical representation of the data flow of a system that can be understood by both technical and non-technical users.

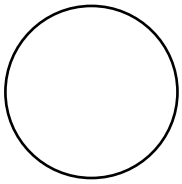
The flow of data in a system or process is represented by a Data Flow Diagram (DFD). It also gives insight into the inputs and outputs of each entity and the process itself. Data Flow Diagram (DFD) does not have a control flow and no loops or decision rules are present. Specific operations, depending on the type of data, can be explained by a flowchart. It is a graphical tool, useful for communicating with users, managers and other personnel. it is useful for analysing existing as well as proposed systems.

It provides an overview of

- What data is system processes?
- What transformation are performed?
- What data are stored?
- What results are produced, etc.

Like all the best diagrams and charts, a DFD can often visually “say” things that would be hard to explain in words, and they work for both technical and nontechnical audiences, from developer to CEO. That’s why DFDs remain so popular after all these years. While they work well for data flow software and systems, they are less applicable nowadays to visualizing interactive, real-time or database-oriented software or systems.

Data Flow Diagram Notations:

Name	Symbol	Description
Process		The tasks performed on the data is known as process. Process is represented by circle. Somewhere round edge rectangles are also used to represent process.
External Entity		These are the starting and ending points for the data flow in a DFD. External entities are placed on the edges of a DFD to represent the input and output of information to the entire system or process.
Data Flow		A data flow symbol represents the data flow occurring between two processes or between, an external entity and a process in the direction of the flow arrow.
Data Store		A data store represents a logical file. The direction of data flow arrow shows whether data is being read from or written into a data store.
Output		The output symbol is used when a hard copy is produced Such as report.

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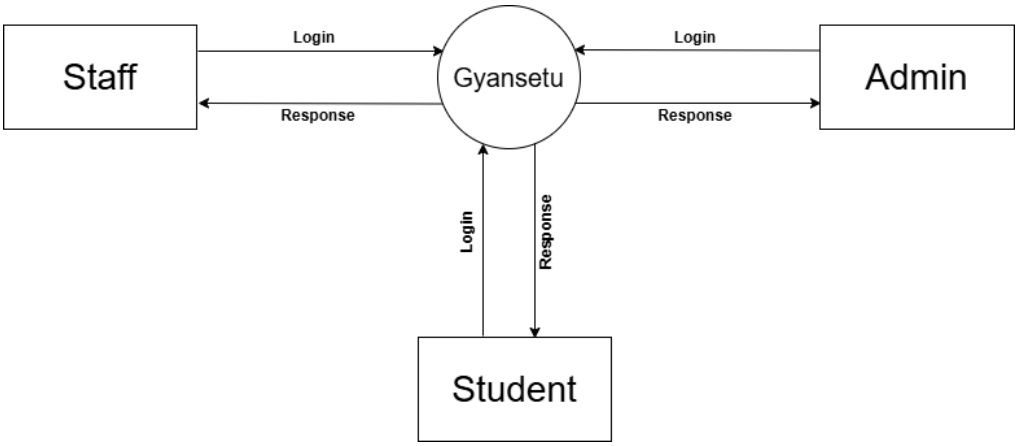
3.1 Context Level DFD – 0 Level:

Level 0 is also called a Context Diagram. It's a basic overview of the whole system or process being analyzed or modelled.

It's designed to be an at-a-glance view, showing the system as a single high-level process, with its relationship to external entities. It should be easily understood by a wide audience, including stakeholders, business analysts, data analysts and developers.



3.2 1st Level DFD:

In 1st level DFD, a context diagram is decomposed into multiple bubbles/process. In this level, we highlight the main objectives of the system and breakdown the high-level process of 0-level DFD into subprocesses.

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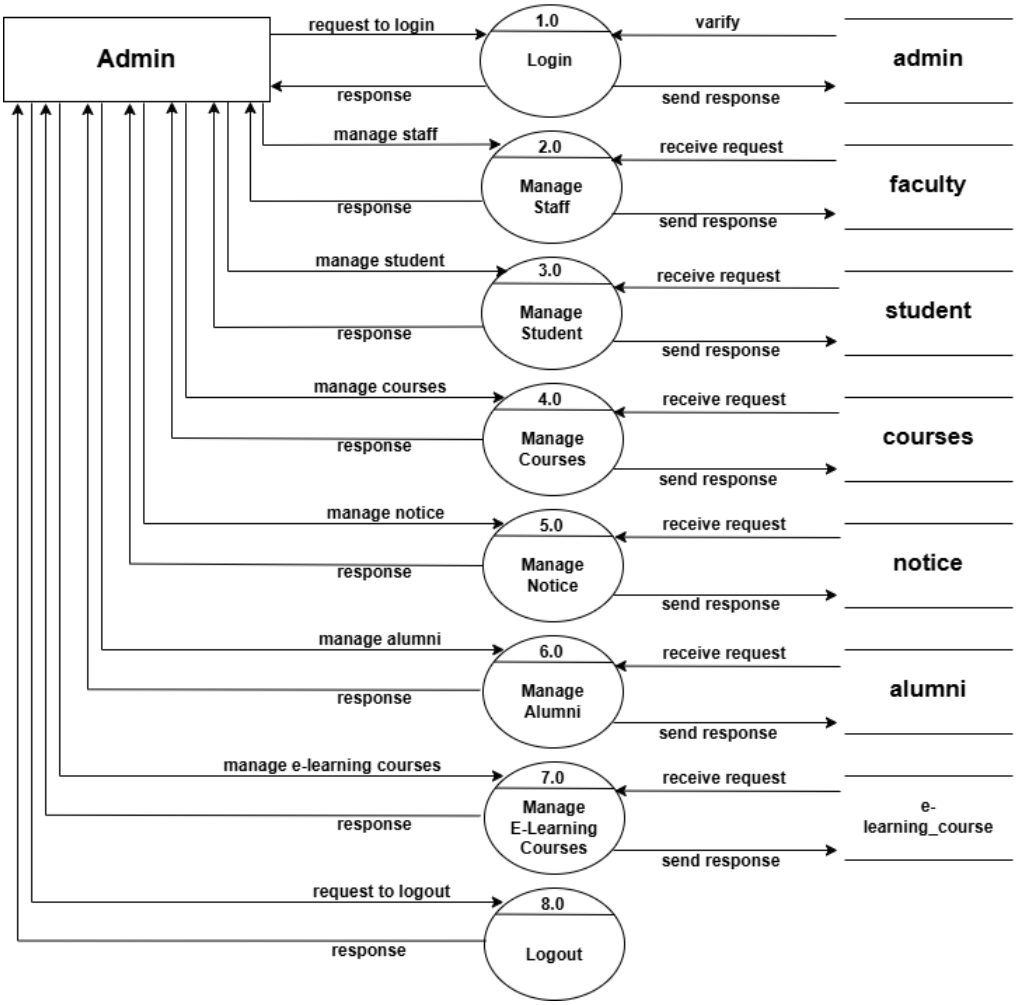
3.2.1 1st Level DFD for Admin:



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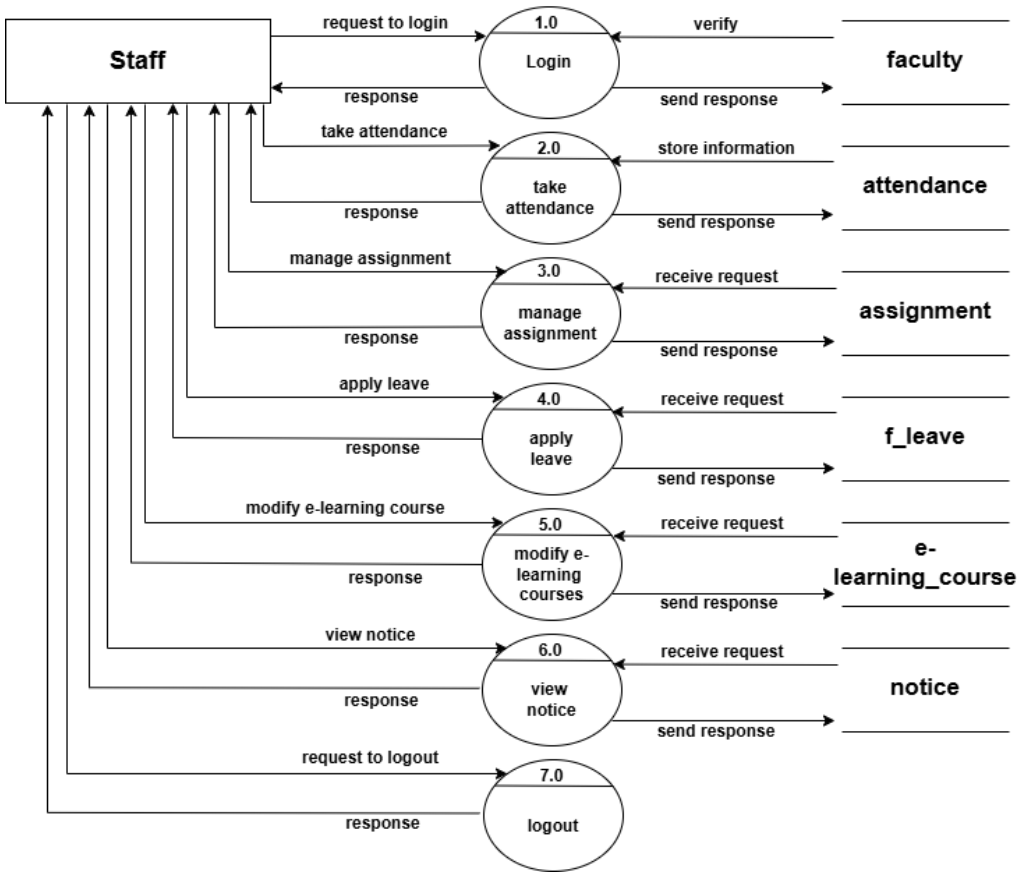
3.2.2 1st Level DFD for Staff:



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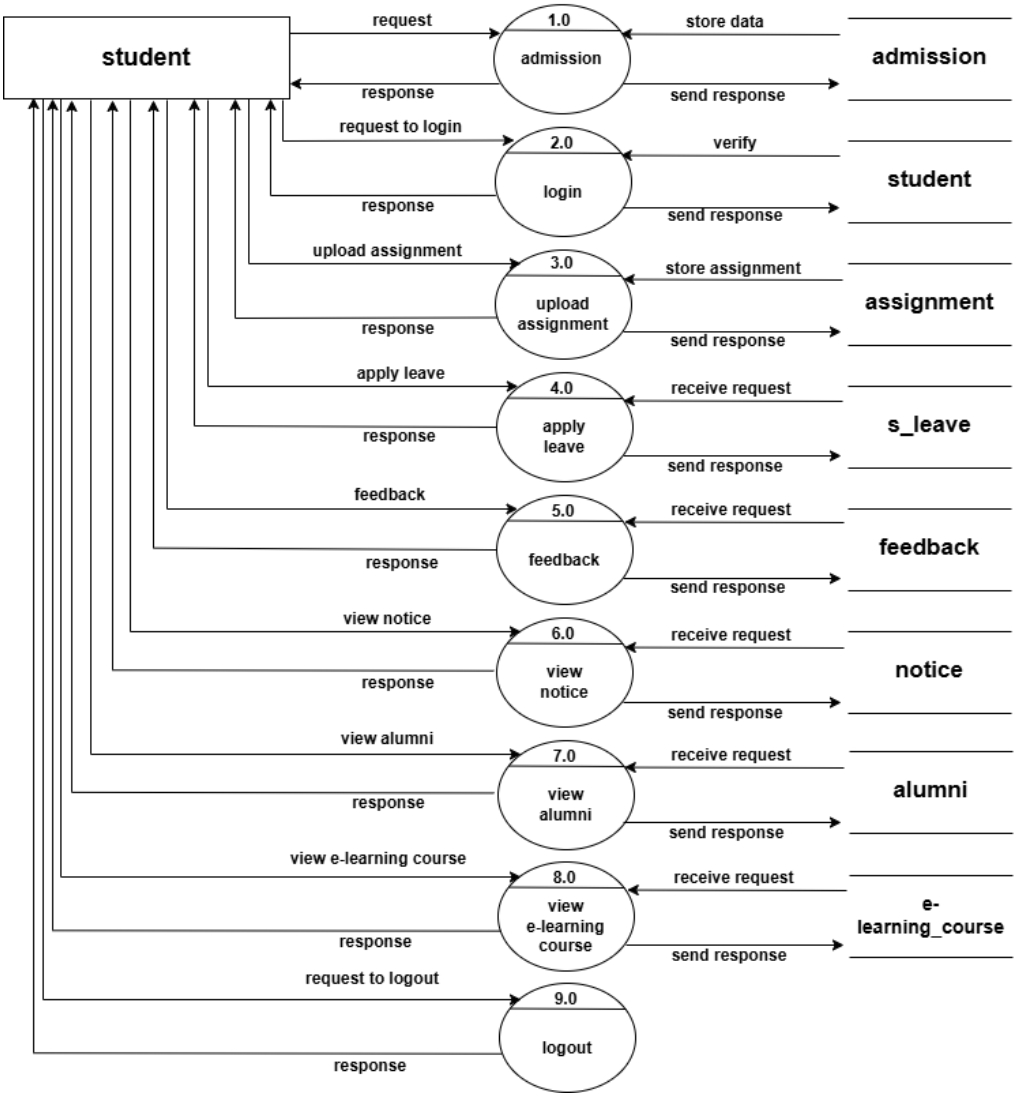
3.2.3 1st Level DFD for Student:



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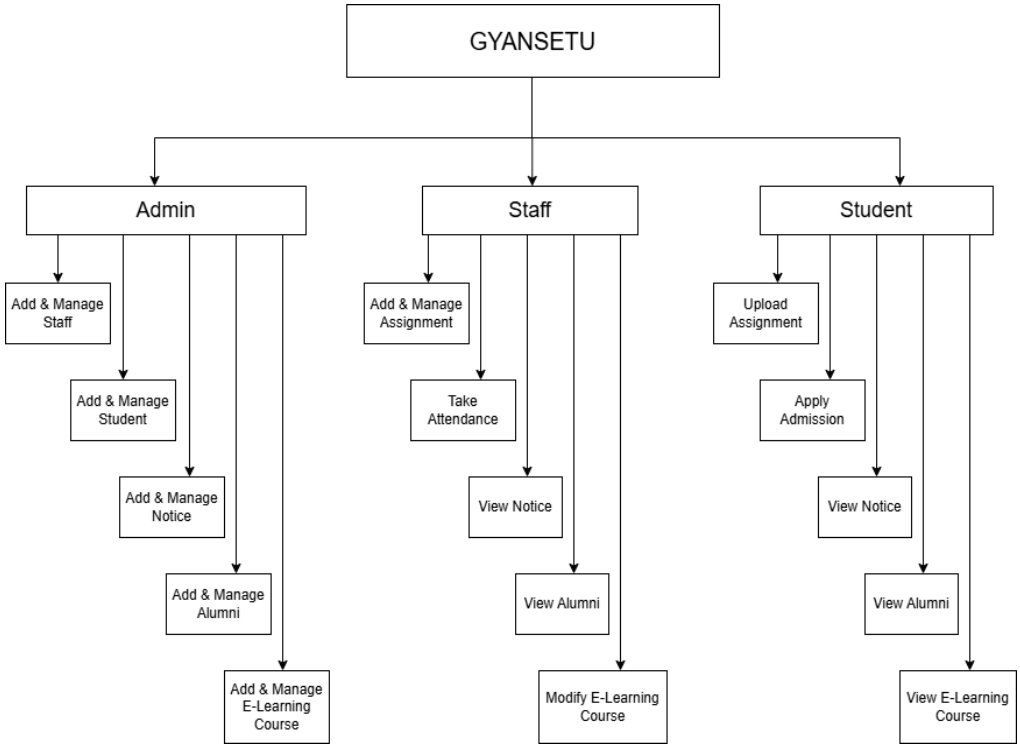
4. ORGANIZATION CHART:



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